CONCURRENT SERVER APPLICATION

Objective:

Here the objective is to know how to handle multiple clients with single server with help of **SOCKET** programming by using **TCP** protocols.

TASK:

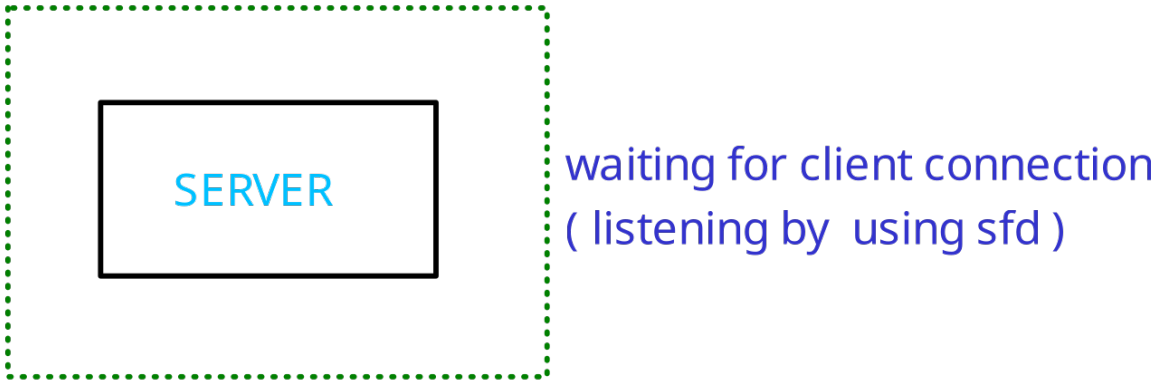
Here multiple clients will try to communicate with server and server should be able to accept connection from every client and should able to send / recv msg.

There are multiple versions can be posible i.e , by using **fork()** , by using **threads** , by using **select()** and **etc... .**

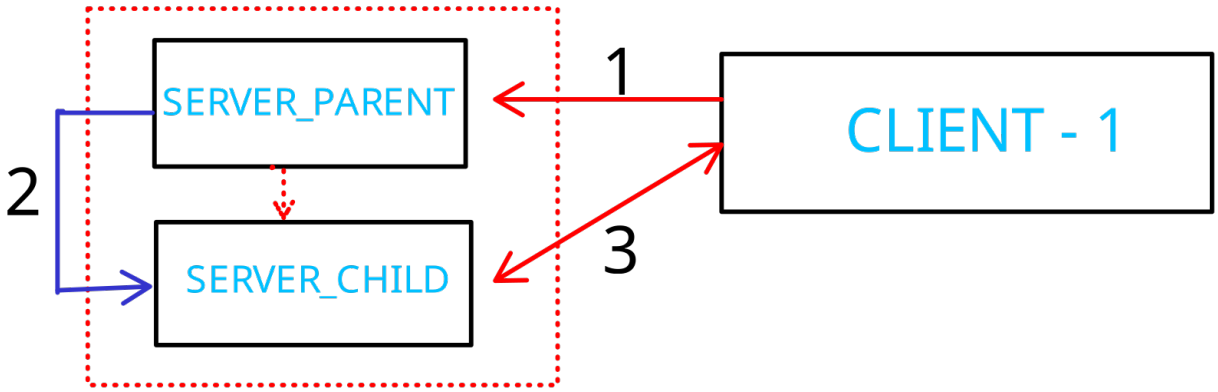
Consider , initially will implement by using fork()

Approuch:

- **First Server will wait at accept() call to accept connection from client1.**



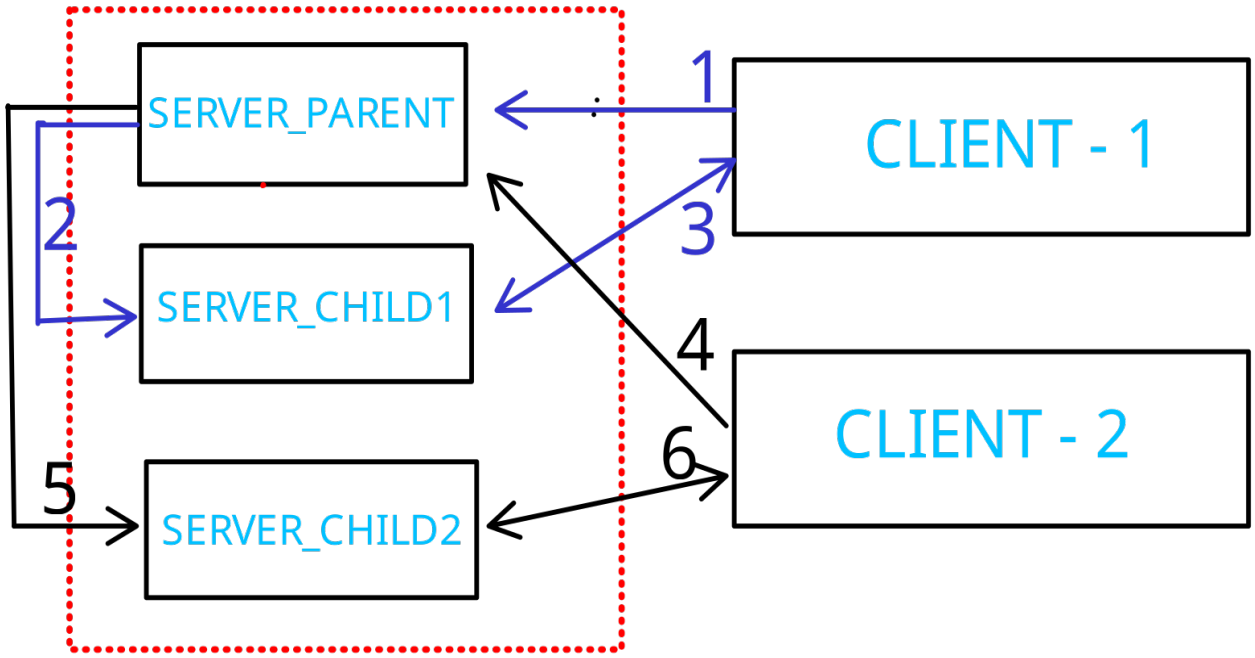
➢ **Now client1 has been launched**



1. **Client1** requested for connection.
2. Server accepted connection , created new file descriptor for data exchange and created one child process to communicate with client1 i.e.., **server_child1.**
3. **Server_child1** will be able to exchange data with only client1.

And then **server_parent** will be waiting for next client.

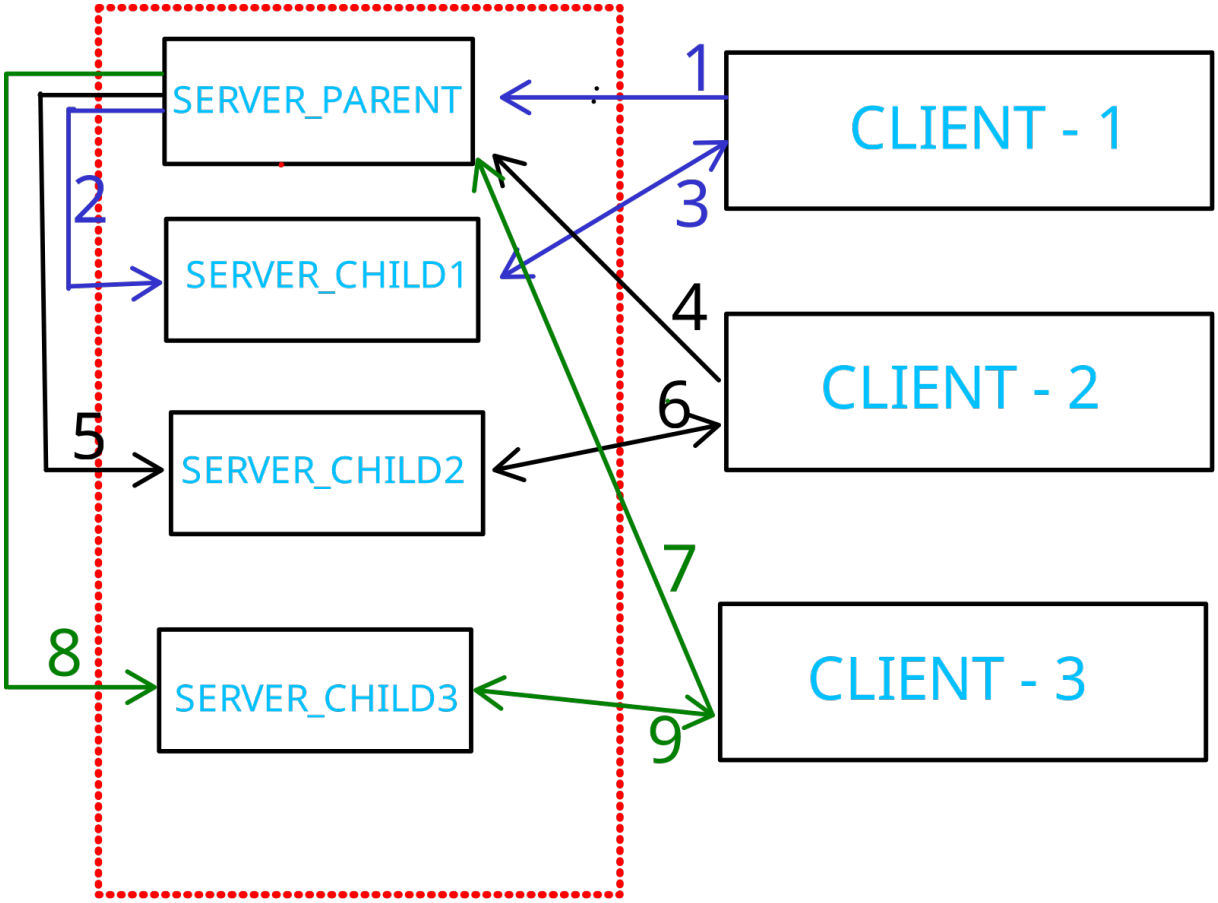
➢ **Now client2 has been launched**



4. **Client2** requested server_parent for connection.
5. Server_parent accepted connection, created new file descriptor for data exchange and created one more child process i.e.., **server_child2.**
6. **Server_child2** only able to exchange data with client2

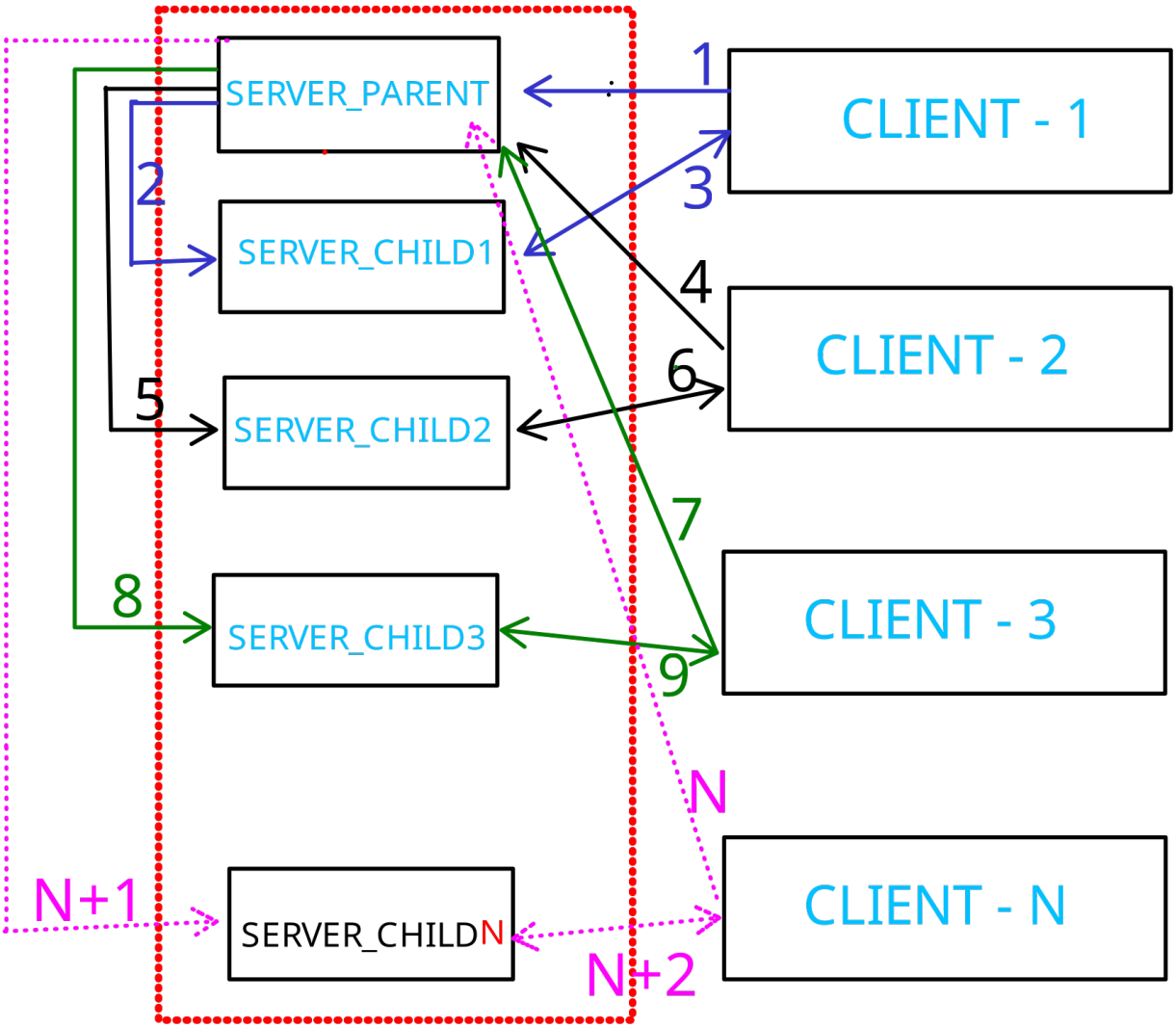
And then **server_parent** will be waiting for next client.

- **Now client3 has been launched**



7. **Client3** requested server_parent for connection.
8. Server_parent accepted connection from client3 and created new file descriptor will be creatred for data exchange and new child createad i.e.., **server_child3**
9. **Server_child3** only able to exchange data with client3

➢ **Now clientN has been launched**



N. ClientN requested for connection with server
N+1. Server_parent accepted connection from clientN , created new file descriptor and creatred one more child for server to handle clientN.
N+2. Server_childN able to exchange data with clientN.

THANK YOU